

Towards Wearable-Free Exergame for Orthostatic Dysregulation: Improving Motion Detection with Smartphone-Only Calibration

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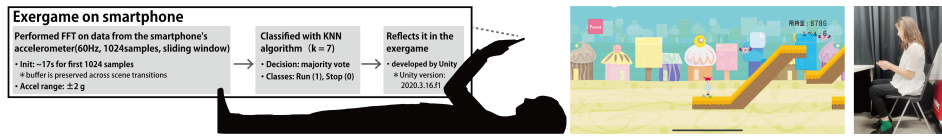


Figure 1: Proposed exergame

Abstract

This study develops a smartphone-only exergame for adolescents with orthostatic dysregulation to reduce sensor burden while maintaining detection accuracy. Furthermore, this study compared the exergame with a conventional method using an external sensor. The results showed that if calibration is tailored to each participant properly, motion detection can be achieved with the same accuracy as that achieved using a sensor.

CCS Concepts

• Human-centered computing → Human computer interaction (HCI).

Keywords

exergame, serious game, orthostatic dysregulation

ACM Reference Format:

Hitomi Miyazaki, Saki Sakaguchi, Tetsuaki Baba, and Kumiko Kushiya. 2025. Towards Wearable-Free Exergame for Orthostatic Dysregulation: Improving Motion Detection with Smartphone-Only Calibration. In *SIGGRAPH Asia 2025 Posters (SA Posters '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3757374.3771438>

1 Introduction

In Japan, severe and prolonged orthostatic dysregulation (OD), an autonomic disorder causing dizziness and palpitations upon standing, are increasing among adolescents, leading to increased

school absenteeism. Although, exercise therapy is effective; Yanagimoto [Yanagimoto 2021] proposed leg exercises in lying and sitting positions but noted difficulty in sustaining patients' motivation. In previous study [Miyazaki et al. 2024], we developed an exergame using leg movements (pulling, paddling) in lying or sitting positions to motivate children with OD. This game is to be used daily at home by OD patients using smartphones and M5StickC sensor. However, sensors like M5StickC require attachment and pairing, which limits daily use. Several methods have been proposed for detecting movements and actions using only smartphones [Hu et al. 2022; Rungrapanak et al. 2018; Xia et al. 2024], but often require preparation, such as using dedicated equipment to mount the smartphone or attaching smartphone to a user's body. Even methods that require minimal setup involve larger detection motions, detect large actions like falls—unsuitable for OD exergames requiring smaller leg movements while holding the device and watching the screen.

Therefore, this study proposes a smartphone-only foot movement detection method compatible with the exergame for OD, and compares its accuracy to traditional sensor-based methods.

2 Design and Implementation

The system diagram is shown in Figure 1. Using the accelerometer built into the smartphone, an FFT was performed on the values obtained from the accelerometer. Considering the ease of implementation in Unity, the analyzed values were classified using kNN to distinguish between moving and stationary states of the foot. A feature vector $\mathbf{x} \in \mathbb{R}^{20}$ was constructed as follows. Let the acceleration components on the x , y , and z axes at the i -th sample be denoted by a_{xi} , a_{yi} , and a_{zi} , respectively. The norm $\|\mathbf{a}_i\|$ at each sample i was computed as:

$$\|\mathbf{a}_i\| = \sqrt{a_{xi}^2 + a_{yi}^2 + a_{zi}^2}, \quad i = 1, \dots, 1024.$$

An FFT was applied to $\|\mathbf{a}_i\|$, and the amplitudes of the first 20 frequency components—mainly representing low-frequency motions,

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SA Posters '25, Hong Kong, Hong Kong

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ACM ISBN 979-8-4007-2134-2/25/12

<https://doi.org/10.1145/3757374.3771438>

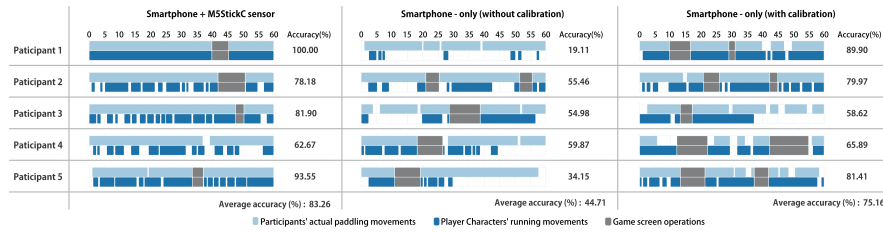


Figure 2: Timeline graph of the movements of participants determined manually from recorded videos and the movements of player character reflected in the exergame (left: with sensor; center: smartphone only without calibration; right: smartphone only with calibration)

which tend to capture human movement were extracted as features:

$$\mathbf{m} = [|m_0|, |m_1|, \dots, |m_{19}|],$$

where m_k is the k -th FFT coefficient. For classification, kNN with $k = 7$ was employed using these vectors to distinguish between two motion classes: “Run” (label 1) and “Stop” (label 0). This classification result is reflected in real-time within the game. In the game, paddling the legs while lying or sitting makes the character run, and tapping the screen triggers jumping and attacking. Gameplay begins with selecting default or new kNN training data (reference samples size: 400 samples, each a 20-dimensional vector). If new, paddling and stopping are recorded for player-specific calibration and saved for reuse. Subsequently, the player selects a stage and starts the 2D side-scrolling game, where the player character run, jump to avoid obstacles, and attack enemies with bullets.

3 Experiment

We compared our smartphone-only method with the M5StickC sensor method in terms of matching accuracy with actual movements. Five participants in their twenties (3 males and 2 females) played the game in a sitting position to avoid posture effects. An iPhone 12 Pro (iOS 15.11) was used in the experiment. Gameplay and screen footage were recorded to assess how accurately in-game player character movements reflected real movements. In the first experiment, participants played the game for a minute each, using an M5StickC sensor and using only the smartphone developed, without the sensor. Participants 1, 3, and 5 played without the sensor first, and Participant 2 and Participant 4 played with the sensor first. At this time, calibration to create kNN learning data for each participant was not performed, and participants used the default learning data. Consequently, the motion detection accuracy was low when not using the sensor. Therefore, a second experiment was conducted in which participants played the same game for 1 min after calibration.

The results of the first experiment are shown on left side and center of Fig.2. The movements of player character in the game within 1 min and the actual movements performed by the participant, which were determined by the recorded video, are shown on a timeline. The accuracy rate is noted at the right of each timeline. When using the sensor, the average accuracy rate was 83.26%, whereas only smartphone without calibration was 44.71%. A paired t-test was conducted on these results, yielding a p-value of 0.0056, indicating a statistically significant difference at the 5% significance level. The results of the second experiment are shown on the right

side of Fig.2. The average accuracy rate was 75.16%. Compared to the uncalibrated condition, and a paired t-test was conducted, yielding a p-value of 0.015, indicating a statistically significant difference at the 5% significance level. Additionally, compared to the with sensor case, a paired t-test was conducted, yielding a p-value of 0.1726, indicating no statistically significant difference at the 5% significance level.

The experiments showed that if calibration is properly conducted, motion detection can be achieved with the same accuracy as that achieved using a sensor. However, this experiment had a few participants, and therefore the results should be interpreted as preliminary. A larger-scale experiment with more participants is necessary to obtain more accurate results. In addition, it is necessary to consider the effort required for calibration, even though no sensor attachment is needed. In future, we will consider the detection of multiple types of motion and designs that reduce the effort of calibration.

Acknowledgments

This work was supported by JST SPRING, Grant Number JPMJSP2156, MIYAKO-MIRAI Project of Tokyo Metropolitan University.

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